

## STAPLES, MN, USA

WMO: 727514

Lat: 46.381N

Lon: 94.806W

Elev: 392

StdP: 96.70

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54910

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -28.9      | -26.3 | -32.3                         | 0.2 | -28.4 | -28.8 | 0.3 | -25.7 | 11.4                  | -8.5 | 10.2 | -8.8 | 1.9                   | 270  | 0.584 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>11.9            | (c)<br>31.1     | (d)<br>23.5 | (e)<br>28.9 | (f)<br>22.4 | (g)<br>27.6 | (h)<br>21.0 | (i)<br>25.3         | (j)<br>29.1 | (k)<br>23.9 | (l)<br>27.5 | (m)<br>22.7 | (n)<br>26.0 | (o)<br>4.7           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.9                            | 19.7 | 27.6 | 22.7 | 18.2 | 26.1 | 22.0 | 17.5 | 25.3 | 79.5          | 29.1 | 74.2 | 27.6 | 68.7 | 26.7 | 30.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.0              | 8.6  | 7.7 | -33.1                      | 33.5 | 3.6                | 1.7 | -35.7                                              | 34.7 | -37.8      | 35.7 | -39.9      | 36.6 | -42.5      | 37.9 |
|                   |      |     | -33.2                      | 26.8 | 3.6                | 1.8 | -35.8                                              | 28.1 | -37.9      | 29.1 | -39.9      | 30.1 | -42.6      | 31.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 5.3    | -12.0 | -10.6 | -2.5  | 5.5   | 12.5  | 17.9  | 20.9  | 19.2  | 14.7  | 6.8   | -0.8  | -9.1  |  |  |
|                                                                        | DBStd       | 12.86  | 7.80  | 7.29  | 7.39  | 5.75  | 4.75  | 3.59  | 3.17  | 3.16  | 4.72  | 4.92  | 6.24  | 7.11  |  |  |
|                                                                        | HDD10.0     | 2886   | 681   | 578   | 391   | 154   | 27    | 1     | 0     | 0     | 11    | 123   | 327   | 593   |  |  |
|                                                                        | HDD18.3     | 4971   | 939   | 811   | 647   | 385   | 190   | 49    | 12    | 27    | 127   | 357   | 575   | 852   |  |  |
|                                                                        | CDD10.0     | 1169   | 0     | 0     | 3     | 20    | 104   | 238   | 337   | 286   | 153   | 26    | 2     | 0     |  |  |
|                                                                        | CDD18.3     | 211    | 0     | 0     | 0     | 1     | 9     | 37    | 91    | 54    | 19    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2071   | 0     | 0     | 0     | 11    | 131   | 368   | 872   | 503   | 174   | 13    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 549    | 0     | 0     | 0     | 2     | 36    | 82    | 259   | 131   | 38    | 2     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.1    | 3.2   | 3.2   | 3.3   | 3.6   | 3.4   | 2.9   | 2.5   | 2.4   | 2.9   | 3.3   | 3.3   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 695    | 18    | 16    | 35    | 57    | 88    | 109   | 102   | 79    | 71    | 67    | 33    | 20    |  |  |
|                                                                        | PrecMax     | 873    | 54    | 42    | 88    | 138   | 178   | 181   | 228   | 174   | 176   | 151   | 104   | 49    |  |  |
|                                                                        | PrecMin     | 510    | 1     | 3     | 11    | 22    | 29    | 38    | 41    | 22    | 18    | 9     | 4     | 7     |  |  |
|                                                                        | PrecStd     | 90     | 12    | 9     | 19    | 25    | 37    | 38    | 43    | 39    | 33    | 41    | 25    | 10    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 6.1   | 8.0   | 18.7  | 24.1  | 31.0  | 31.3  | 33.0  | 32.4  | 30.2  | 25.9  | 17.8  | 6.4   |  |  |
|                                                                        |             | MCWB   | 3.6   | 5.9   | 13.3  | 15.8  | 19.9  | 23.5  | 25.5  | 24.4  | 21.9  | 16.9  | 12.7  | 4.0   |  |  |
|                                                                        | 2%          | DB     | 2.3   | 4.8   | 14.0  | 21.1  | 27.2  | 28.8  | 31.2  | 29.8  | 27.5  | 21.2  | 13.9  | 3.0   |  |  |
|                                                                        |             | MCWB   | 1.1   | 2.8   | 9.5   | 13.2  | 17.9  | 22.6  | 24.3  | 22.6  | 20.8  | 15.0  | 9.8   | 1.9   |  |  |
|                                                                        | 5%          | DB     | 0.9   | 2.4   | 11.0  | 18.0  | 23.8  | 27.3  | 29.0  | 27.8  | 25.9  | 17.6  | 11.1  | 2.0   |  |  |
|                                                                        |             | MCWB   | 0.3   | 1.2   | 8.3   | 11.1  | 16.2  | 20.3  | 23.0  | 21.4  | 19.7  | 12.8  | 7.8   | 1.3   |  |  |
|                                                                        | 10%         | DB     | -1.3  | 0.0   | 7.3   | 16.0  | 22.0  | 25.9  | 27.7  | 26.3  | 22.8  | 15.0  | 7.7   | 0.0   |  |  |
|                                                                        |             | MCWB   | -1.5  | -0.5  | 5.1   | 9.7   | 15.3  | 19.2  | 21.7  | 20.4  | 18.3  | 11.1  | 5.8   | -0.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.7   | 6.4   | 14.3  | 17.7  | 22.6  | 25.7  | 27.7  | 26.8  | 24.3  | 19.3  | 14.3  | 4.6   |  |  |
|                                                                        |             | MCDB   | 5.8   | 8.2   | 17.4  | 24.0  | 26.7  | 29.5  | 30.9  | 29.4  | 28.5  | 22.4  | 16.3  | 5.7   |  |  |
|                                                                        | 2%          | WB     | 1.2   | 3.1   | 10.8  | 14.3  | 19.6  | 23.9  | 25.9  | 24.2  | 22.6  | 16.0  | 10.4  | 2.5   |  |  |
|                                                                        |             | MCDB   | 2.0   | 4.5   | 12.4  | 19.0  | 24.4  | 27.4  | 29.7  | 27.8  | 25.7  | 20.0  | 13.8  | 3.4   |  |  |
|                                                                        | 5%          | WB     | -0.1  | 1.2   | 7.7   | 12.1  | 17.6  | 22.2  | 24.3  | 22.9  | 20.9  | 13.7  | 7.8   | 1.1   |  |  |
|                                                                        |             | MCDB   | 0.5   | 2.1   | 10.0  | 17.0  | 21.4  | 25.3  | 28.0  | 26.2  | 23.9  | 17.0  | 9.9   | 1.7   |  |  |
|                                                                        | 10%         | WB     | -2.0  | -0.9  | 5.1   | 10.2  | 15.9  | 20.3  | 23.0  | 21.6  | 19.2  | 11.8  | 5.8   | -0.6  |  |  |
|                                                                        |             | MCDB   | -1.8  | -0.1  | 7.3   | 15.2  | 20.2  | 24.0  | 26.3  | 24.8  | 22.1  | 14.5  | 8.0   | -0.1  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 10.0  | 11.0  | 10.5  | 11.5  | 12.0  | 11.4  | 11.9  | 11.9  | 11.6  | 10.0  | 9.0   | 8.6   |  |  |
|                                                                        | 5% DB       | MCDBR  | 9.8   | 11.2  | 14.1  | 16.4  | 16.4  | 13.8  | 13.4  | 13.8  | 14.4  | 14.5  | 13.7  | 9.9   |  |  |
|                                                                        |             | MCWBR  | 8.5   | 9.1   | 9.6   | 9.7   | 8.8   | 7.3   | 7.2   | 7.4   | 8.1   | 8.9   | 9.2   | 8.3   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.5   | 10.3  | 13.0  | 14.7  | 13.6  | 11.5  | 12.2  | 12.0  | 11.8  | 13.0  | 12.4  | 9.2   |  |  |
|                                                                        |             | MCWBR  | 8.3   | 8.6   | 9.2   | 9.4   | 8.5   | 7.7   | 7.6   | 7.4   | 7.7   | 8.9   | 9.0   | 8.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.260 | 0.294 | 0.307 | 0.358 | 0.379 | 0.379 | 0.381 | 0.385 | 0.354 | 0.311 | 0.282 | 0.260 |  |  |
|                                                                        | taud        |        | 2.374 | 2.276 | 2.381 | 2.324 | 2.328 | 2.362 | 2.371 | 2.370 | 2.437 | 2.541 | 2.511 | 2.411 |  |  |
|                                                                        | Ebn at Noon |        | 850   | 878   | 923   | 898   | 887   | 886   | 879   | 860   | 861   | 859   | 824   | 810   |  |  |
|                                                                        | Edh at Noon |        | 71    | 97    | 102   | 119   | 123   | 119   | 117   | 113   | 96    | 75    | 63    | 61    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.62  | 2.74  | 3.83  | 4.83  | 5.38  | 5.90  | 6.27  | 5.27  | 3.95  | 2.45  | 1.52  | 1.12  |  |  |
|                                                                        | RadStd      |        | 0.19  | 0.29  | 0.38  | 0.40  | 0.40  | 0.39  | 0.26  | 0.46  | 0.30  | 0.38  | 0.15  | 0.14  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | -2.59   | -2.78  | N/A     | -2.67 | -2.86 | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (51 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page